

Project Report: AbhiHomes - House Hunt App

1. INTRODUCTION

1.1 Project Overview

AbhiHomes is a comprehensive real estate rental platform built using the MERN stack. It allows users to browse, filter, and submit property listings while integrating a seamless Google login experience. The app provides a smooth user experience with a dark-themed modern UI and animated transitions.

1.2 Purpose

The purpose of AbhiHomes is to streamline the rental process by providing a user-friendly platform where renters can find homes and owners can submit listings.

2. IDEATION PHASE

2.1 Problem Statement

Renters and property owners often face difficulties in finding and listing rental properties. There is a need for a unified platform that simplifies this process.

2.2 Empathy Map Canvas

Empathy map involves understanding users—renters and property owners—what they see, feel, hear, and do in the rental process.

2.3 Brainstorming

Ideas such as integrating Google login, map-based search, and owner approval mechanisms were explored.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

From visiting the homepage to submitting or booking a rental property, user flow was mapped to optimize experience.

3.2 Solution Requirement

The solution requires authentication, property listing, filtering, submission forms, admin approval, and dashboards.

3.3 Data Flow Diagram

Describes data flow between frontend (React), backend (Express.js), and database (MongoDB).

3.4 Technology Stack

Frontend: React.js + Tailwind CSS + Vite

Backend: Node.js + Express.js + MongoDB + Mongoose

Auth: Google OAuth

Deployment: Vercel (Frontend), Render (Backend)

4. PROJECT DESIGN

4.1 Problem Solution Fit

A direct, easy-to-use interface solves the listing/search issue for both renters and owners.

4.2 Proposed Solution

An app that authenticates users, allows home submissions and filtered searches, and provides role-based dashboards.

4.3 Solution Architecture

Client-server model with frontend (React) making API calls to backend (Express.js), using MongoDB for persistence.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Planned using Agile sprints: Ideation, Design, Development, Testing, and Deployment phases.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Tested for login speed, API response times, and rendering performance using tools like Postman and browser dev tools.

7. RESULTS

7.1 Output Screenshots

Screenshots included: Homepage, Search Filter, Property Submission Form, and Dashboard.

8. ADVANTAGES & DISADVANTAGES

Advantages:

- Fast search with filters
- Secure login via Google
- Responsive and modern UI

Disadvantages:

- Relies on third-party services
- Admin approval not fully automated

9. CONCLUSION

AbhiHomes successfully provides a modern solution for house hunting and rental submissions with a user-centric interface.

10. FUTURE SCOPE

Potential to integrate payment systems, GPS-based recommendations, chat between renters and owners, and mobile app development.

11. APPENDIX

Source Code: https://github.com/abhiramvasishta/ABHI_HOMES-HOUSEHUNT-

Dataset Link: Not applicable

GitHub & Project Demo Link: https://github.com/abhiramvasishta/ABHI_HOMES-HOUSEHUNT-